

SEBASTIAN CARABALLO

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EDUCATION

Purdue University

B.S. in Mechanical Engineering

West Lafayette, IN

Exp. May 2028

- **Honors:** Dean's List and Semester Honors ('24), Alpha Lambda Delta & Phi Eta Sigma Honor Societies
- **Coursework:** Object-Oriented Programming, Linear Algebra, Data Science, Multivariable Calculus, Statistics, Diff. Equations

SKILLS

- **Languages:** Python, JavaScript/TypeScript (React, Node.js), SQL, MATLAB, C++, Java, HTML/CSS
- **Software:** Next.js, FastAPI, PostgreSQL, Git, Vercel, ArduPilot, AutoCAD, SolidWorks, Siemens NX, Docker, Tailwind, Figma
- **Equipment:** Wiring, Soldering, DAC/ADC boards, Laser Cutting, 3D Printing, CNC, GD&T, Raspberry Pi, Sensors, Arduino

EXPERIENCE

FitScript | fitscript.io

Founder / CEO

Miami, FL

May 2026 - Present

- Founded FitScript, an AI nutrition coach for diabetic and GLP-1 users, and built the full-stack web app solo in Next.js, TypeScript, and PostgreSQL: an LLM coach with persistent memory and condition-aware meal logging.
- Selected a diabetic-first market with a custom segment-scoring tool that ranked 3 customer segments against a locked 6-criteria rubric. Grew a 64-person waitlist ahead of a fall 2026 private beta.

Algoma

Software Engineering Intern

Brooklyn, NY

June 2026 - August 2026

- Returned for a second summer in a deeper engineering role. Shipped production features and UX fixes to the React property-discovery platform through Jira-ticketed pull requests and code review.
- Built a configuration panel for AI-generated site renders, letting users set render style, camera angle, and surrounding context before generation.

Product & Software Intern

May 2025 - August 2025

- Designed and programmed interactive models for a client-facing analytics page, enabling analysis of up to 10 filtered properties from hundreds. Client feedback in 8+ demos highlighted the improved interface.
- Implemented custom data validation and weighting algorithms to analyze complex variables, cutting bad-input errors and improving system performance in production.

Vlachos Research Group | Cardio Fabrication & Flow Loop Teams

Undergraduate Research Assistant

West Lafayette, IN

January 2026 - Present

- Led system bring-up of a fluidic testbed, interfacing DAC/ADC boards with high-fidelity sensors. Wrote automated Python workflows that synchronize high-speed imaging with real-time sensor data.
- Designed and fabricated precision casting molds and 0.3x-scale models with additive manufacturing. Ran producibility reviews to find bottlenecks and turn models into manufacturable hardware.

Purdue National Security & Defense Society (NSDS)

UAV Engineer, Unmanned Systems

West Lafayette, IN

December 2025 - August 2026

- Built a hexacopter that places self-sufficient Meshtastic radio repeaters on water towers to restore communications after natural disasters, developed with Jefferson County, FL emergency response.
- Owned drone construction, the payload deployment mechanism, and ArduPilot integration. Produced mechanical Tooling Build-to-Packages (BTP) and engineering drawings to meet strict structural durability requirements.

Purdue University | First-Year Engineering

Undergraduate Teaching Assistant, ENGR 131

West Lafayette, IN

September 2026 - Present

- Support student teams through design projects and Python programming in Purdue's first-year engineering course. Grade against rubrics with written feedback and run weekly help sessions.

PROJECTS

Earnings Trader | [Live Demo](#)

Personal Project | LLM Trading-Signal and Backtesting Engine

Python, FastAPI, React

May 2026 - Present

- Built an LLM pipeline that reads SEC EDGAR 8-K filings and scores earnings language for sentiment, guidance quality, and risk flags. Generated 50+ trade signals across 10 tickers with GPT-4o-mini.
- Engineered a vectorbt backtesting engine measuring 34 trades against an SPY benchmark on Sharpe ratio, win rate, and max drawdown. Deployed a live FastAPI and React dashboard to publish results.

AWARDS/PUBLICATIONS

- **Publications:** Skomra A., Russon A., Chevli K.K., Caraballo S. "Combination of Pre-Biopsy ExoDx Testing and mpMRI Improves Predictive Power of High-Grade Prostate Cancer." AUA 2023 Annual Meeting abstract, Journal of Urology 209(4S), 2023.
- **Awards:** FRC Regional Awards: Industrial Design ('22), Creativity ('23), Engineering Inspiration ('23)
- **Involvement:** PURPL Turbojet Sub-Team (2024-25); American Society of Mechanical Engineers (ASME); Theta Chi Fraternity